Chirag Venkatesh

Residence: Bangalore, India
E-mail: chirag42012@gmail.com ✻ *Telephone:* +91-8762503230
Title: Electronics & Communication Engineer

Technical Skills

Physical Design	**RTL2GDSII Flow**, Synthesis, Floorplanning, Placement Optimization, CTS, Routing, Timing Closure, **MCMM Analysis**
Timing Analysis	**STA**, Setup/Hold Analysis, **WNS/TNS Optimization**, **SDC Constraints**, Parasitic Extraction Concepts
EDA Tools	Cadence Virtuoso, Xilinx Vivado (Conceptual exposure to ICC2, Innovus, PrimeTime, Tempus)
Languages	**TCL**, **Python**, C, MATLAB
Automation	Flow scripting, Timing report parsing, Constraint validation, Makefile-based builds
Physical Verification	DRC/LVS Concepts, Layout vs Schematic checks
Core Concepts	**CMOS VLSI Design**, Digital IC Design, Clock Tree Design, Signal Integrity basics

Professional Experience

External Consultant — *Dec 2024 – Aug 2025*
Hardware Engineer — *Bangalore, India*

- Performed timing and performance analysis on embedded hardware systems, applying concepts aligned with **STA**, including **setup/hold checks** and **latency optimization**.
- Debugged timing-related failures using structured log analysis and constraint validation techniques, aligned with **SDC constraint debugging** and **timing closure** workflows.
- Developed automation scripts in **Python** and **TCL** for validation, timing report parsing, and workflow optimization, improving debugging efficiency and turnaround time.
- Worked in multi-configuration hardware environments, ensuring deterministic timing behavior and optimizing system-level latency—aligned with **MCMM** analysis.
- Contributed to **GAN charger** hardware manufacturing and testing, focusing on **performance validation**, **reliability**, and timing consistency across operating conditions.
- Applied foundational knowledge of **digital design** and **CMOS VLSI**, including **signal integrity** considerations and clock behavior awareness in high-speed systems.

Bharat Earth Movers Ltd. (Rail & Metro) — *Aug 2024 – Nov 2024*
Software Engineer — *Bangalore, India*

- Worked on **Train Control and Monitoring System (TCMS)**, integrating multiple onboard subsystems with focus on **deterministic communication** and real-time behavior.
- Developed and validated modules over the **CAN communication network**, ensuring reliable data transfer, **low-latency signaling**, and timing consistency across distributed nodes.
- Collaborated with hardware teams to analyze and maintain system-level timing integrity, aligned with **setup/hold constraints** and **timing synchronization**.
- Optimized communication workflows for **deterministic latency** and performance, analogous to **timing closure** and path optimization.
- Contributed to validation and verification emphasizing **robustness**, **fault tolerance**, and timing stability across operating conditions (aligned with **MCMM** scenarios).

Projects

Readimentary (RSVP PDF Reader)

- Built a browser-based RSVP reader with **ORP-centered word rendering** and adjustable WPM for focus-optimized reading.
- Implemented chapter detection and persistent local library using **IndexedDB** and **localStorage**.
- Developed navigation controls with progress memory, jump-to-section, and chapter resume logic.
- Added UI state management for pause/resume, reading history, and interactive session controls.
- Delivered a full in-browser flow with **React 19**, **Vite 7**, **Tailwind CSS**, and **PDF.js**.

Sneaki — AI-Powered Activity to Google Calendar

- Built an automated system that logs application/window activity into Google Calendar with AI-based task categorization.
- Processes webhook activity payloads and extracts task name, category, and productivity score via JSON-returning AI backends.
- Creates structured calendar events with computed durations and metadata (app name, URL, task, score).
- Supports local or hosted AI endpoints and resilient handling of missing data.

Statistical Aerboreal Bird Repeller

- Built a computer vision pipeline using **SIFT** and **HOG** descriptors for robust bird detection under scale and lighting changes.
- Compared live video frames against a labeled dataset of bird species images to compute similarity scores.
- Designed a density estimator that aggregates detections over time to produce stable control signals.
- Mapped density scores to **PWM duty cycles** for dynamic ultrasonic actuation instead of fixed bursts.
- Documented results and field testing in a **peer-reviewed conference paper**.

Real-Time Gesture-Controlled Robotic Hand

- Built a mechanical system mirroring hand movements in **real time** with low-latency feedback.
- Implemented **Python-based signal processing** for sensor filtering, smoothing, and gesture stability.
- Used **MQTT** messaging to stream and synchronize servo state updates across devices.
- Tuned motor response curves to reduce jitter and improve fine-grained positional accuracy.

Education

PES University — *Bangalore, India*
B.Tech in Electronics and Communication Engineering — *2020 – 2024*
Minor in Computer Science Engineering